MDM

Subject :- ME24052 – Introduction to Robotics & Automation

Topic:- Robot Drive System and Sensors



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Unit-2: Robot Drive Systems & Sensors	[08 Hrs.]

Drive Systems :- Pneumatic Drives, Hydraulic Drives, Mechanical Drives, Electrical Drives-D.C. Servo Motors, Stepper Motors, A.C. Servo Motors, BLDC-Salient Features, Applications and Comparison of all these Drives, Micro actuators, selection of drive, Power transmission systems for robot, Motion conversion (Gearbox -Planetary, Harmonic, Cycloidal gearbox), VFD.

Sensors:- Transducers and sensors, Sensors in robotics, Principles and applications of the following types of sensors- Proximity Sensors, Photoelectric Sensors, Position sensors – Piezoelectric Sensor, LVDT, Resolvers, Encoders – Absolute and Incremental: - Optical, Magnetic, Capacitive, pneumatic Position Sensors, Range Sensors- Range Finders, Laser Range Meters, Touch Sensors, Force and torque sensors

Robot Drive System

- The actions of the individual joints must be controlled in order for the manipulator to perform a desired movement of its body, arm, motion and wrist. This provided by the drive system used to power the robot.
- The joints are moved by actuators powered by a particular form of drive system.
- Common drive systems used in robotics are electric drive, hydraulic drive, and pneumatic drive.

- Types of Actuators

  1**. Hydraulic actuators**

  2. Pneumatic actuators

  3. Electric Motors, like: Servomotors, Stepper motors or Direct-drive electric motors

Mechanical Drive Systems

- The drive system determines the speed of the arm movement, the strength of the robot, dynamic performance, and, to some extent, the kinds of application.



- A robot will require a *drive system* for moving their arm, wrist, and body. A drive system is usually used to determine the capacity of a robot. For actuating the robot joints, there are *three different types* of drive systems available such as:

(i) Electric drive system,

(ii) Hydraulic drive system, and

(iii) Pneumatic drive system.

- The most importantly used two types of drive systems are electric and hydraulic.

Electric Drive System:

- The electric drive systems are capable of moving robots with *high power* or speed. The actuation of this type of robot can be done by either DC servo motors or DC stepping motors. It can be well –suited for rotational joints and as well as linear joints. The electric drive system will be perfect for *small robots* and precise applications. Most importantly, it has got greater accuracy and repeatability. The one disadvantage of this system is that it is slightly costlier. An example for this type of drive system is *Maker 110 robot*.

Hydraulic Drive System:

- The hydraulic drive systems are completely meant for the *large –sized robots*. It can deliver high power or speed than the electric drive systems. This drive system can be used for both linear and rotational joints. The rotary motions are provided by the rotary vane actuators, while the linear motions are produced by hydraulic pistons. The *leakage* of hydraulic oils is considered as the major disadvantage of this drive. An example for the hydraulic drive system is *Unimate 2000 series robot.*



Pneumatic Drive System:

- The pneumatic drive systems are especially used for the *small type robots*, which have less than five degrees of freedom. It has the ability to offer fine accuracy and speed. This drive system can produce rotary movements by actuating the rotary actuators. The translational movements of sliding joints can also be provided by operating the piston. The price of this system is *less* when compared to the hydraulic drive. The drawback of this system is that it will not be a perfect selection for the *faster operations.*

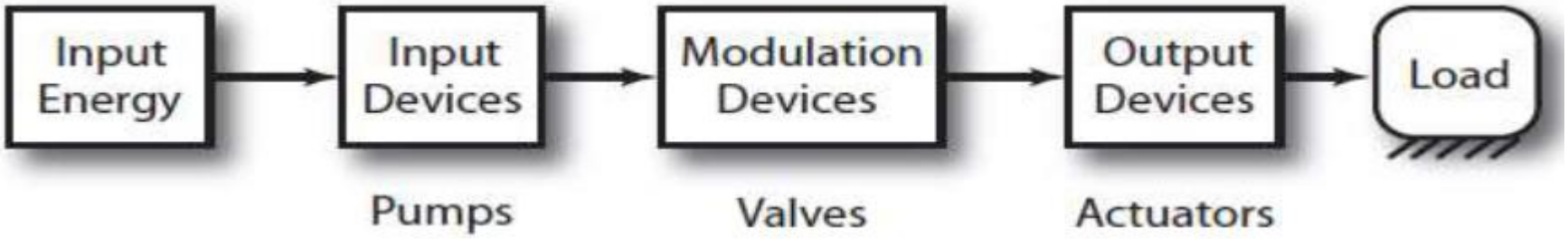
Pneumatic - Hydraulic Drive System:

Introduction to Fluid Power

- The basic fluid power system consists of a source of input energy and a suitable device for energy input, energy output, and energy modulation.
- Transmitting fluid power requires a pump to convert the mechanical energy into fluid energy.
- Proper devices are needed for the modulation of fluid actuators. The primary source of input energy is quite often an electric motor, an internal combustion engine, or another type of mechanical device that can supply force and motion to operate the pumps.
- The pump supplies hydraulic fluid or pneumatic pressure to the system.
- In other words, fluid power can be defined as the power transmitted and controlled through the use of a pressurized fluid.

The block diagram for a fluid power control system is shown in following figure. A fluid power system consists of three devices:

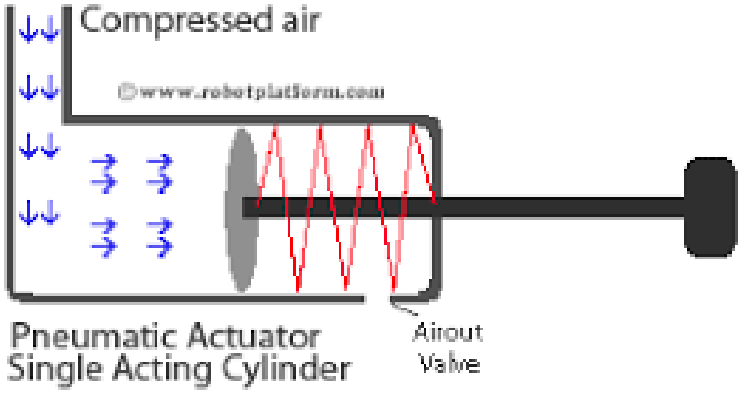
- **Energy input device**
- **Energy modulation device**
- **Energy output device**



Actuator:

- Actuator is a device which provides enough force needed to start the mechatronics systems. At the same time, power should be supplied to the actuator to activate it.
- The power supplied to actuators might be anyone of the following forms such as compressed air, pressurized fluid, electric power and mechanical power.
- If compressed air is supplied to the system, it is called as **pneumatic system**.
- But if pressurized fluid is supplied to flow the system, it is called as **hydraulic system**.





- Pneumatic and hydraulic actuation are two types of fluid power actuation.
- They use pressurized (or compressed) air or liquid (fluids such as hydraulic oil) to transmit power to actuators, which are devices that convert this power into mechanical motion.
- Hence they are widely used in various industries for controlling and moving mechanical components.



Pneumatic Actuation:

- Pneumatic actuation uses compressed air to power actuators.
- The compressed air is typically generated by an air compressor, which is driven by an electric motor.
- The compressed air is then stored in a tank and distributed to the actuators through a network of pipes.
- The principle of operation of a pneumatic actuator is simple.
- When the compressed air enters the actuator, it expands and pushes a piston or diaphragm. The piston or diaphragm is connected to a linkage, which converts the linear motion of the piston or diaphragm into rotational motion.

- In pneumatic systems, force is produced by gas. It is mainly by air pressure acting on the surface of a piston or valve.
- Compressed air is produced in a compressor and stored in a receiver.
- From compressor, it is send to valves which control the direction of fluid flow. Also, flow control valves control the amount of power by the cylinder. The force acting on the piston is given by the equation.
- Force = Pressure X Area = p x A

BASIC PNEUMATIC SYSTEM

- Pneumatic systems use pressurized air to make things move.
- A basic pneumatic system consists of an air generating unit and an air-consuming unit.
- Air compressed in a compressor is not ready for use. The air has to be filtered, the moisture present in air has to be dried, and, for different applications in the plant, the pressure of air has to be varied.
- Several other treatments are given to the air before it reaches finally to the actuator.

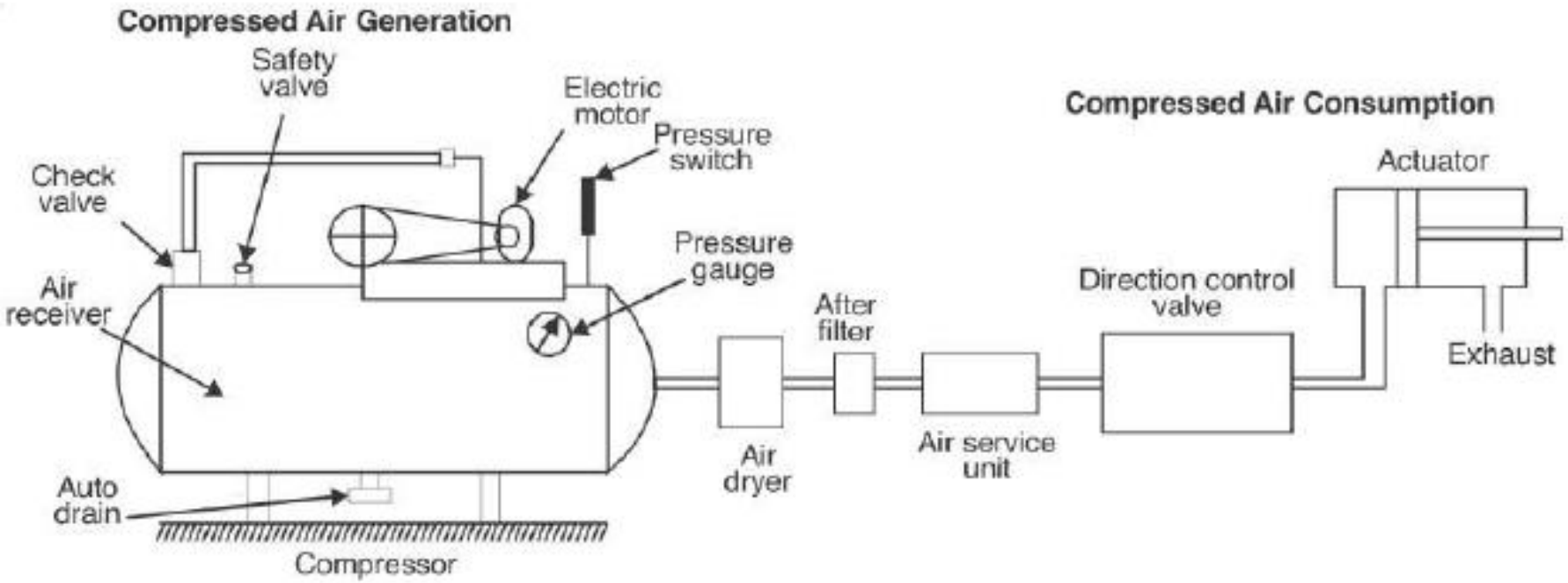


FIGURE 3.2 Basic Pneumatic System.

A typical pneumatic power system includes the following components:

- Compressor
- Electric motor
- Air receiver
- Pressure switch
- Safety valve
- Auto drain
- Check valve
- Pressure gauge
- Air dryer
- After filter
- Air service unit
- Direction control valve
- Pneumatic actuator

- **Compressor:** A device, which converts mechanical force and motion into pneumatic fluid power, is called a compressor. Every compressed air system begins with a compressor, as it is the source of airflow for all the downstream equipment and processes.
- **Electric Motor:** An electric motor is used to drive the compressor.
- **Air Receiver:** An air receiver is a container in which air is stored under pressure.
- **Pressure Switch:** A pressure switch is used to maintain the required pressure in the receiver. It adjusts the *high pressure limit* and *low pressure limit* in the receiver. The compressor is automatically turned off when the pressure is about to exceed the high limit and it is also automatically turned on when the pressure is about to fall below the low limit.
- **Safety Valve:** The function of the safety valve is to release extra pressure if the pressure inside the receiver tends to exceed the safe pressure limit of the receiver.

- **Auto Drain:** Air condenses to give out water in the receiver and a device called auto drain directs this water out.
- **Check Valve:** The valve enables flow in one direction and blocks flow in a counter direction is called check valve. Once compressed air enters the receiver via check valve, it is not allowed to go back even when the compressor is stopped.
- **Pressure Gauge:** The pressure gauge tells us the pressure inside the compressor receiver.
- **Air Dryer:** A device for reducing the moisture content of the working compressed air.
- **After Filter:** A filter that follows the compressed air dryer and usually used for the protection of downstream equipment from desiccant dust, *etc.* is called an after filter. The name filter refers to a device whose primary function is the removal of insoluble contaminants from a liquid or a gas with the help of porous media.
- **Air Service Unit:** Filter, regulator, and lubricator combined in one device is popularly known as an air service unit or F.R.L unit. Its purpose is to supply air to other successive applications in the line. It provides clean air at the required pressure with lubricant added to it to increase the life of equipment.

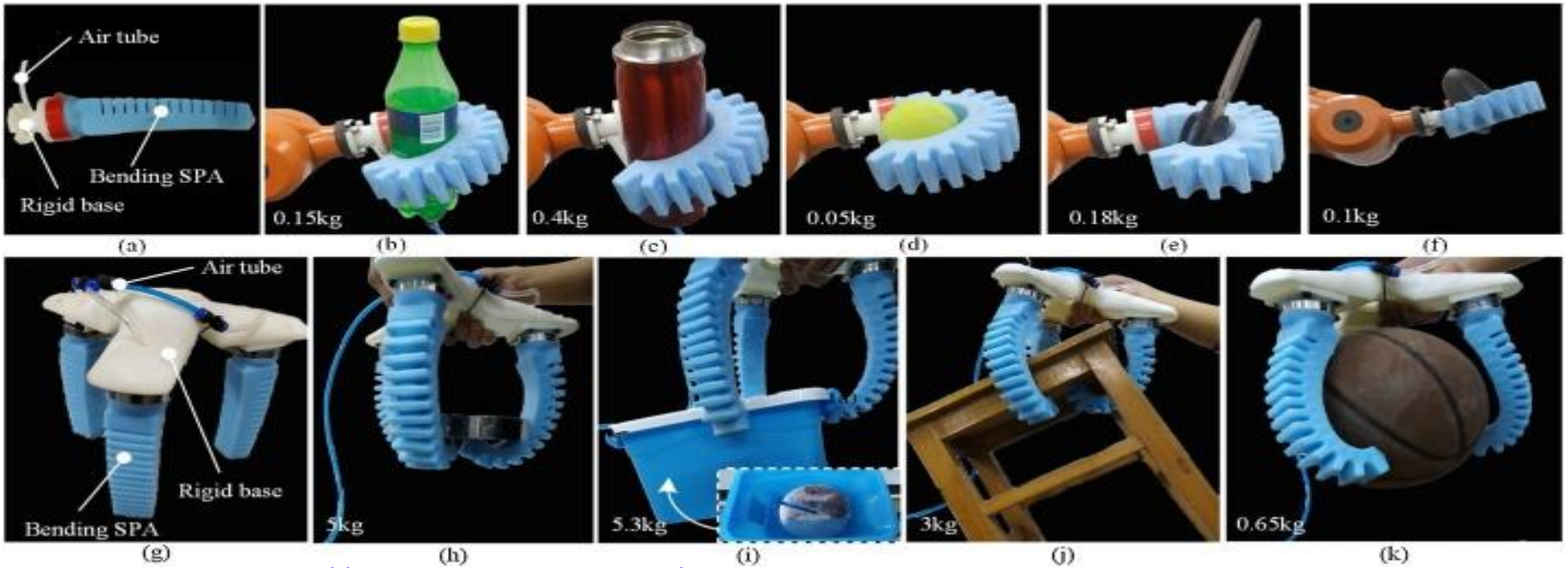
- **Direction Control Valve:** Directional-control valves are devices used to change the flow direction of fluid within a pneumatic/hydraulic circuit. They control compressed-air flow to cylinders, rotary actuators, grippers, and other mechanisms in packaging, handling, assembly, and countless other applications. These valves can be actuated either manually or electrically.
- **Pneumatic Actuator:** A device in which power is transferred from one pressurized medium to another without intensification. Pneumatic actuators are normally used to control processes requiring quick and accurate response, as they do not require a large amount of motive force. They may be reciprocating cylinders, rotating motors, or may be robot end effectors.

A few more components can be added to the system; first, an air intake filter and second, an air intercooler (if a multistage compressor is used). The function of the former is to prevent the entry of vast quantities of dust and dirt along with air and the latter is to cool the air again to room temperature after it is discharged from the low-pressure compressor.

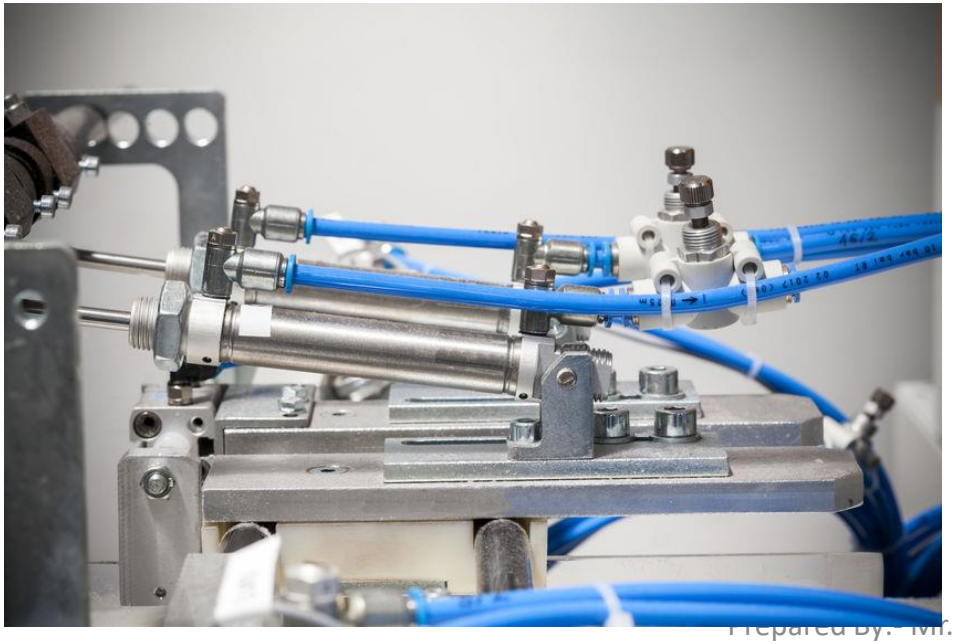
Applications of Pneumatic actuation system

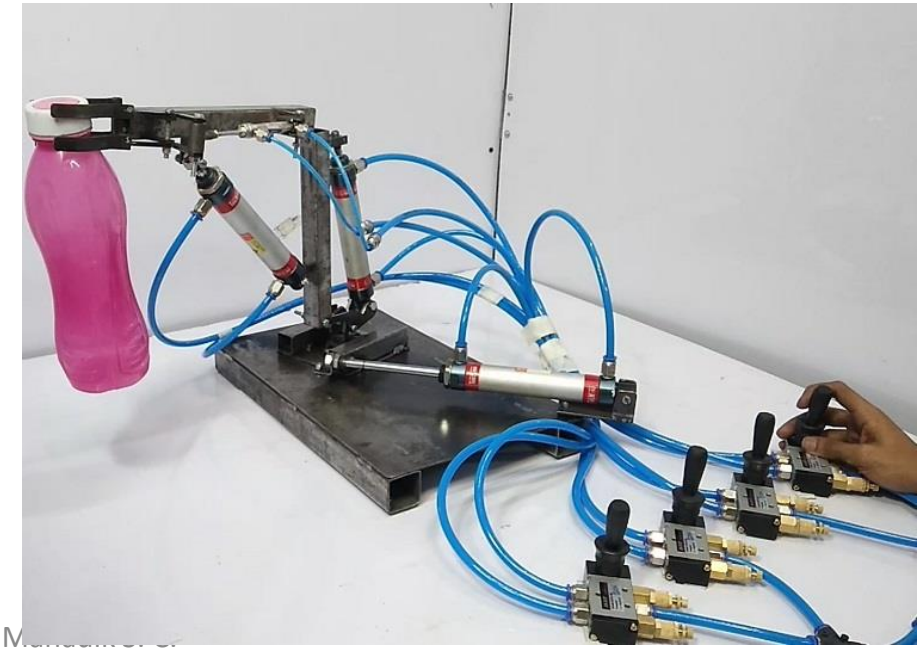
Applications of Pneumatic Actuation: Pneumatic actuation systems find extensive use in various industries due to their advantages, which include simplicity, cleanliness, and ease of control. Some common applications include:

- **Manufacturing**: Pneumatic actuators are used in assembly lines for tasks like clamping, lifting, and pushing.
- **Automotive**: Pneumatic systems are used for brake and clutch actuation, tire **changing** equipment, and suspension systems.



https://www.youtube.com/watch?app=desktop&v=FQKscv_od1w





BASIC HYDRAULIC SYSTEM

A typical hydraulic power system includes the following components:

- Electric motor
- Hydraulic pump
- Strainers and filters
- Pressure gauge
- Pressure relief valves
- Check valve
- Direction control valve
- Hydraulic actuator
- Reservoir

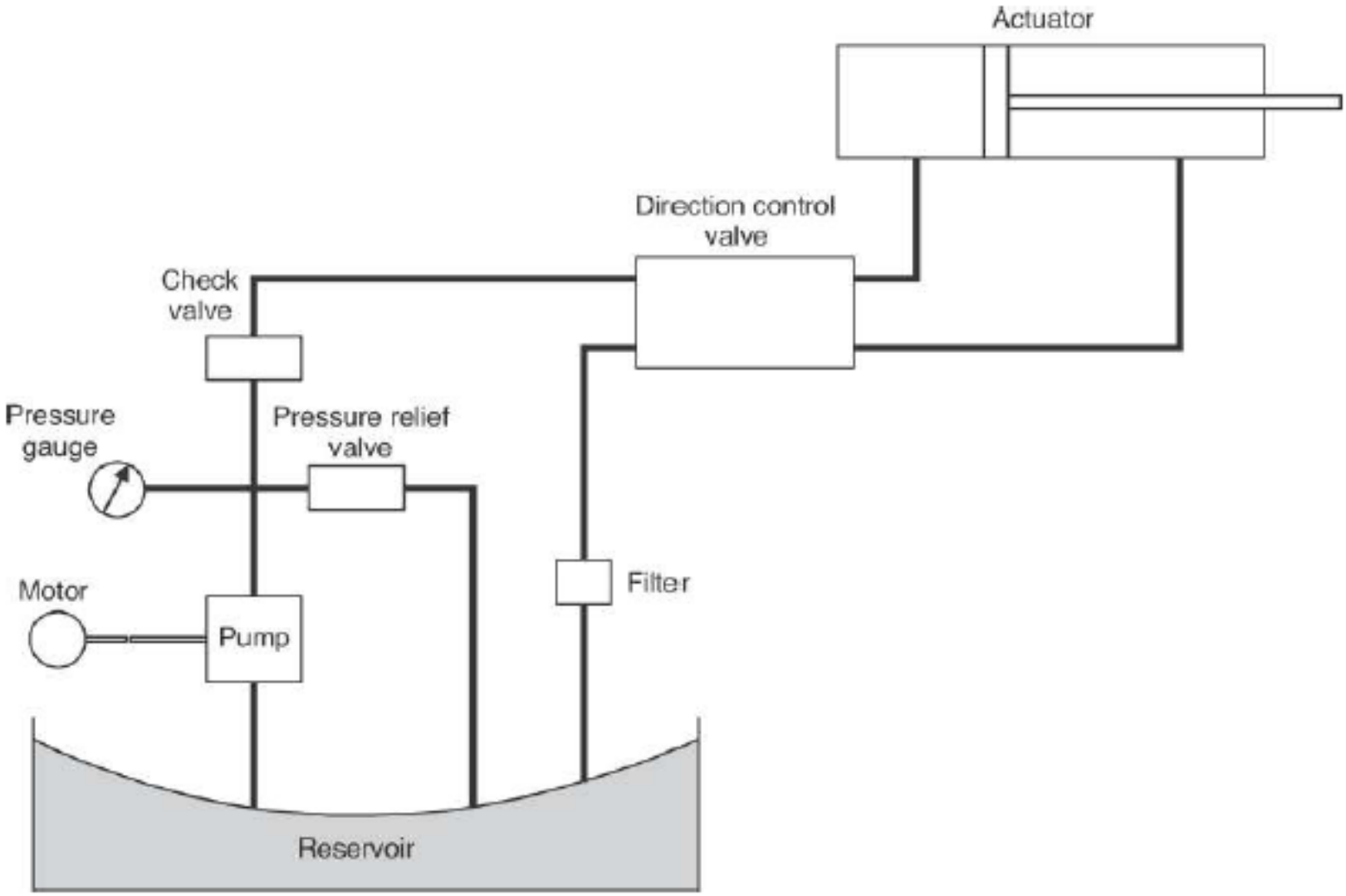


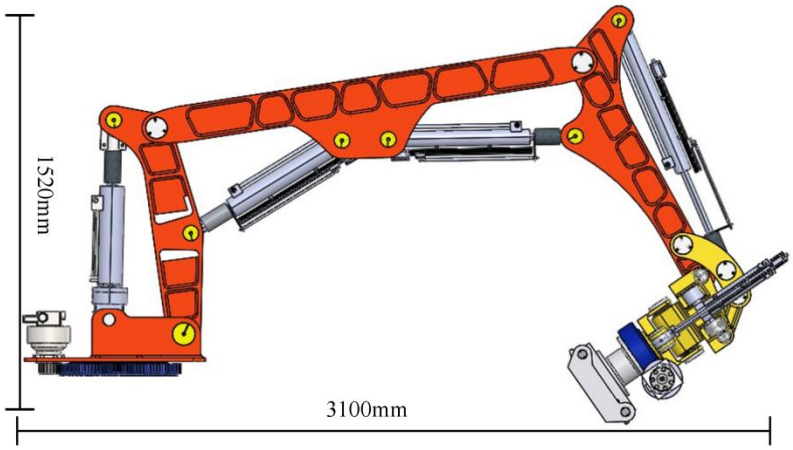
FIGURE 3.3 Basic Hydraulic System.

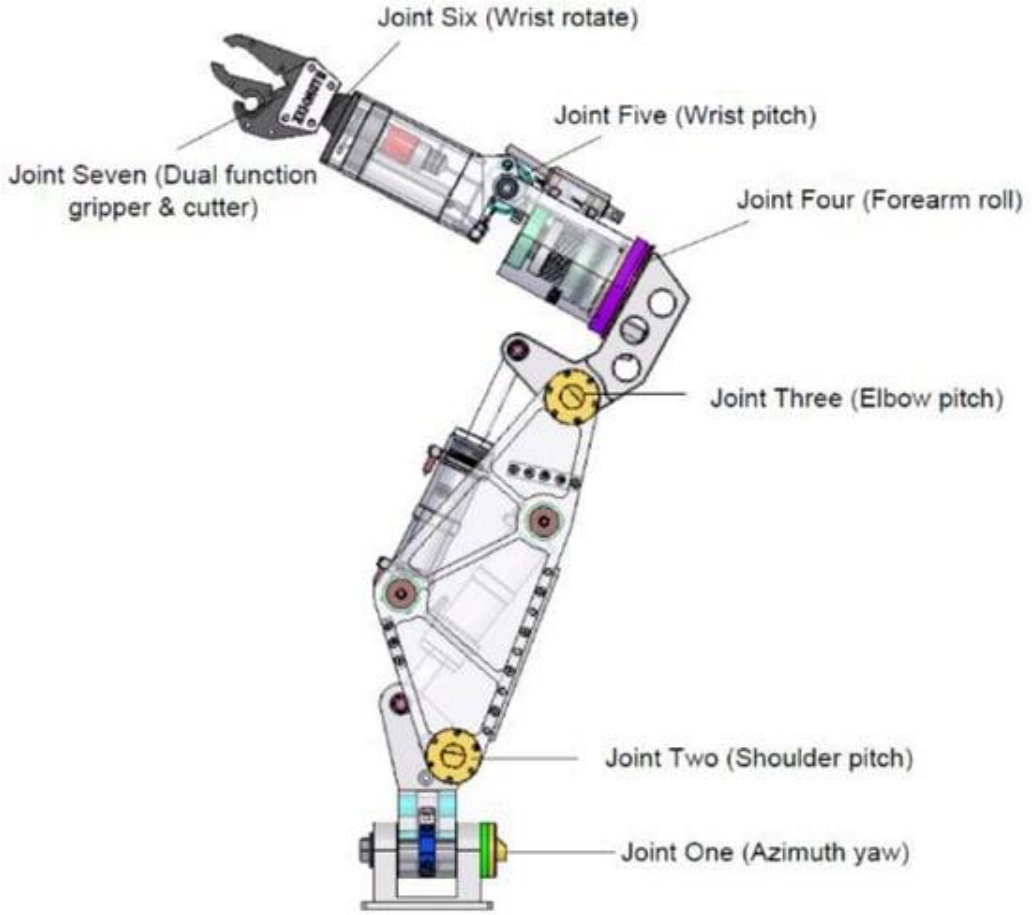
- **Electric Motor:** An electric motor is used to drive the pump.
- **Hydraulic Pump:** Hydraulic pumps convert mechanical energy from a prime mover (engine or electric motor) into hydraulic (pressure) energy. The pressure energy is used then to operate an actuator. Pumps push on a hydraulic fluid and create flow.
- **Strainers and Filters:** To keep hydraulic components performing correctly, the hydraulic liquid must be kept as clean as possible. Foreign matter and tiny metal particles from normal wear of valves, pumps, and other components are going to enter a system. Strainers, filters, and magnetic plugs are used to remove foreign particles from a hydraulic liquid and are effective as safeguards against contamination.
- ***Strainers*:** A strainer is the primary filtering system that removes large particles of foreign matter from a hydraulic liquid. Even though its screening action is not as good as a filter's, a strainer offers less resistance to flow.
- ***Filters*:** A filter removes small foreign particles from a hydraulic fluid and is most effective as a safeguard against contaminates

- **Pressure Gauge:** A pressure gauge tells us the pressure of fluid going into the valve.
- **Pressure Relief Valves:** Relief valves are the most common type of pressure-control valves. The relief valves function may vary, depending on a system's needs. They can provide overload protection for circuit components or limit the force or torque exerted by a linear actuator or rotary motor. The internal design of all relief valves is similar. The valves consist of two sections: a body section containing a piston that is retained on its seat by a spring(s), depending on the model, and a cover or pilot-valve section that hydraulically controls a body piston's movement. The adjusting screw adjusts this control within the range of the valves. Valves that provide emergency overload protection do not operate as often because other valve types are used to load and unload a pump. However, relief valves should be cleaned regularly by reducing their pressure adjustments to flush out any possible sludge deposits that may accumulate. Operating under reduced pressure will clean out sludge deposits and ensure that the valves operate properly after the pressure is adjusted to its prescribed setting.
- **Check Valve:** Check valves are the most commonly used in fluid powered systems. They allow flow in one direction and prevent flow in the other direction. They may be installed independently in a line, or they may be incorporated as an integral part of a sequence, counterbalance, or pressure-reducing valve. The valve element may be a sleeve, cone, ball, poppet, piston, spool, or disc. Force of the moving fluid opens a check valve; backflow, a spring, or gravity closes the valve.

- **Direction Control Valve:** Directional-control valves are devices used to change the flow direction of fluid within a pneumatic/hydraulic circuit. They control compressed-air flow to cylinders, rotary actuators, grippers, and other mechanisms in packaging, handling, assembly, and countless other applications.
- **Hydraulic Actuator:** A hydraulic actuator receives pressure energy and converts it to mechanical force and motion. An actuator can be linear or rotary. A linear actuator gives force and motion outputs in a straight line. It is more commonly called a cylinder but is also referred to as a ram, reciprocating motor, or linear motor. A rotary actuator produces torque and rotating motion. It is more commonly called a hydraulic motor or motor.
- **Reservoir:** A reservoir stores a liquid that is not being used in a hydraulic system.

Applications of Hydraulic Actuation in robotics





Difference Between Hydraulic and Pneumatic Drive

S. No	Hydraulic drive	Pneumatic drive
1.	In hydraulic drive system oil is used as working fluid	In pneumatic drive system air is used as working medium
2.	Hydraulic fluid is costlier	Air is plentiy available in atmosphere. So, it is free
3.	It is suitable for heavy load application	It is suitable for light load application
4.	It has low speed with good speed control	It has high speed and difficult to control speed
5.	It is operate in silent and dirty	It is operate in noisy and clean.

Different Types of Hydraulic and Pneumatic Actuators

1. Single acting Cylinder

- The cylinder is only powered in one direction and needs another force to return it such as an external load or a spring.
- This cylinder only has a head-end port and is operated hydraulically/pneumatically in one direction.
- When pressurized fluid is send into a port, it pushes the plunger/piston rod thus extending it.
- To return or retract a cylinder, fluid must be exhausted out. A plunger returns either because of the weight of a load or from some mechanical force such as a spring.

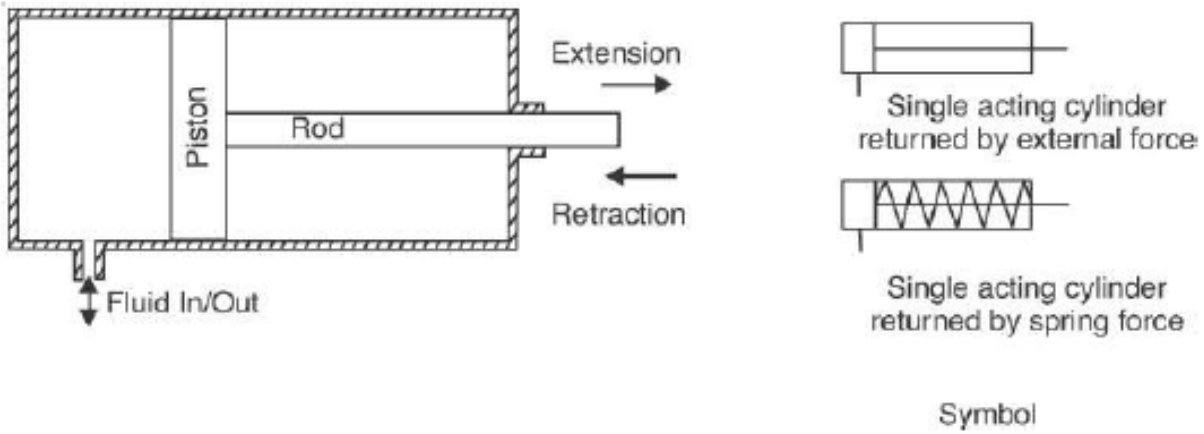
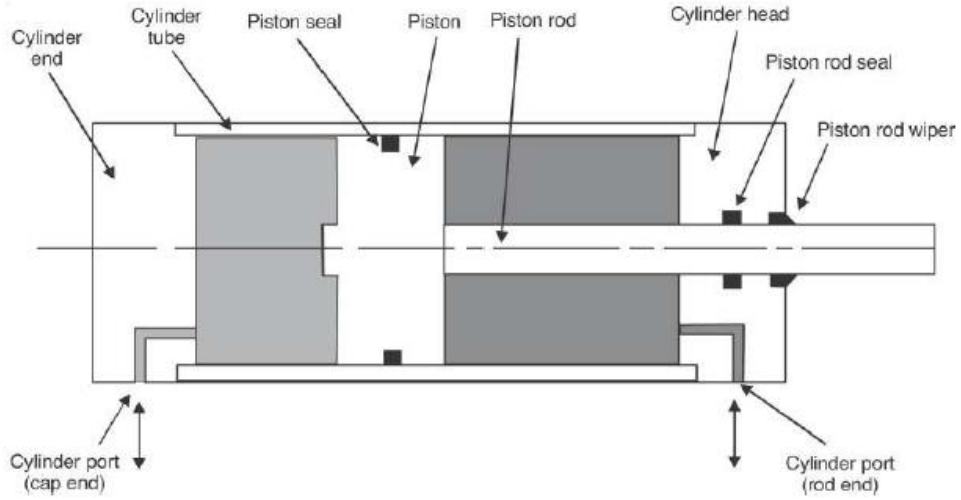


FIGURE 6.3 (a) Single-Acting Cylinder.

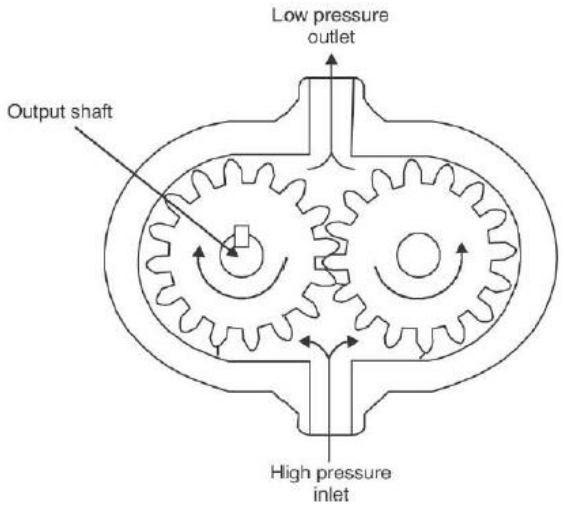
2. DOUBLE-ACTING CYLINDER

- Double-acting cylinders are the most commonly used cylinders in hydraulic and pneumatic applications.
- In this design, pressure can be applied to either side of the cylinder.
- The extension and retraction of cylinder is due to hydraulic/pneumatic pressure.



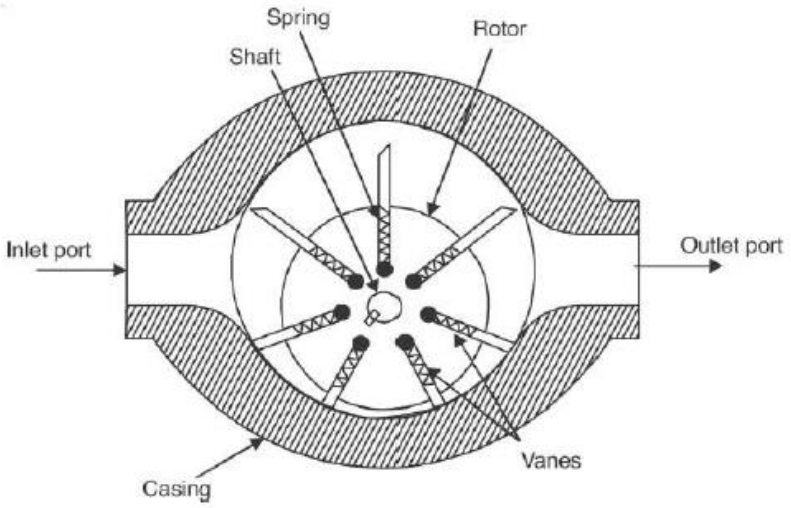
3. GEAR MOTORS

- Gear motors are the most common hydraulic units. These consist of a pair of matched spur or helical gears enclosed in a case.
- Flow from the pump enters the inlet port and flows in either direction around the inside surface of the casing, forcing the gears to rotate as indicated.
- This rotary motion is then available for work at the output shaft.



4. VANE MOTORS

- When pressure oil enters the inlet port, the unequal area of the vanes results in the development of a torque in the motor shaft.
- The larger the exposed area of the vane, or higher the pressure, the more torque is developed, which makes the shaft rotate.
- This particular motor provides rotation in only one direction.



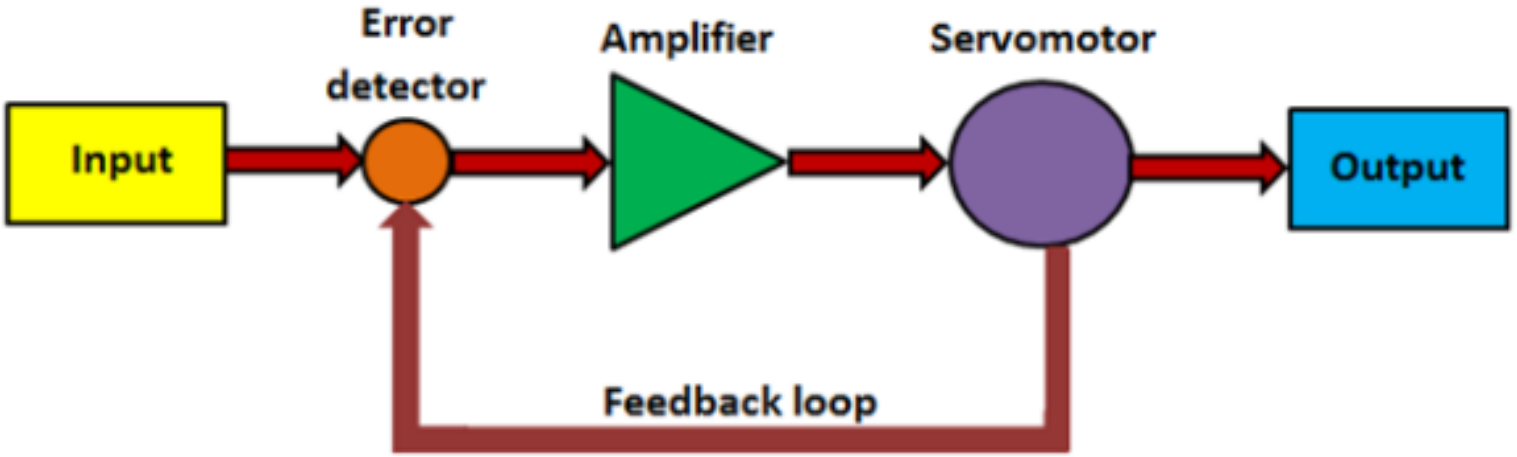
Electrical Drive

- An electrical drive can be defined as an electromechanical device for converting electrical energy into mechanical energy to impart motion to different machines and mechanisms for various kinds of process control.
- Motors are electric devices that convert electrical energy to mechanical motion in the form of a rotor rotating around a stationary axis. These versatile devices are a driving force in a wide range of applications, including Robotics, security cameras, smart locks and 3D printers.

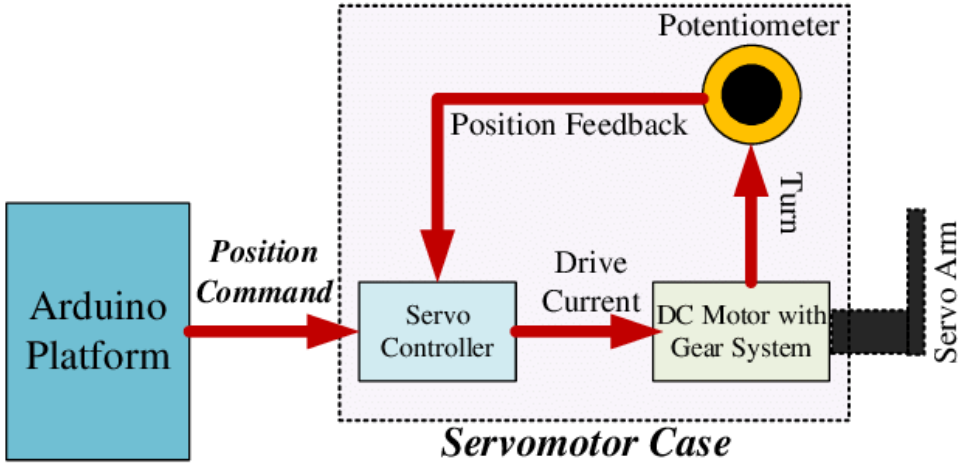
Servomotor

- Servomotors are special electromechanical devices that produce precise degrees of rotation.
- A servo motor is a DC or AC or brushless DC motor combined with a position sensing device.
- Servomotors are also called control motors as they are involved in controlling a mechanical system.
- The servomotors are used in a closed-loop servo system as shown in Figure.
- A reference input is sent to the servo amplifier, which controls the speed of the servomotor.
- A feedback device is mounted on the machine, which is either an encoder or resolver.
- This device changes mechanical motion into electrical signals and is used as a feedback.

- This feedback is sent to the error detector, which compares the actual operation with that of the reference input.
- If there is an error, that error is fed directly to the amplifier, which will be used to make necessary corrections in control action.
- In many servo systems, both velocity and position are monitored.
- Servomotors provide accurate speed, torque, and have ability of direction control.



- **DC servomotors** DC operated servomotors are usually respond to error signal abruptly and accelerate the load quickly.
- A DC servo motor is actually an assembly of four separate components, namely:
    1. DC motor
    2. Gear assembly
    3. Position-sensing device
    4. Control circuit



- **AC servo motor** In this type of motor, the magnetic force is generated by a permanent magnet and current which further produce the torque.
- It has no brushes so there is little noise/vibration.
- This motor provides high precision control with the help of high resolution encoder.
- The stator is composed of a core and a winding.
- The rotor part comprises of shaft, rotor core and a permanent magnet.
- Digital encoder can be of optical or magnetic type.
- It gives digital signals, which are in proportion of rotation of the shaft.

Advantages of servo motors

- Provides high intermittent torque, high torque to inertia ratio, and high speeds
- Work well for velocity control
- Available in all sizes
- Quiet in operation
- Smoother rotation at lower speeds

Disadvantages of servo motors

- More expensive than stepper motors
- Require tuning of control loop parameters
- Not suitable for hazardous environments or in vacuum
- Excessive current can result in partial demagnetization of DC type servo motor

Stepper Motors

- A stepper motor converts electrical pulses into precise mechanical motion. As its name implies, stepper motors operate in discrete steps, where each step is a precise angle of rotation (typically about 1.8°). Stepper motors rotate a set number of times based on how many electrical pulses they have received. Because each rotation follows the exact angle of rotation, stepper motors are highly controllable.
- The main components of a stepper motor are:
- **Rotor:** The rotor (connected to the shaft) is the rotating component in a stepper motor. The rotor has teeth or magnetic poles that produce motion when they interact with the stator.
- **Stator:** The stator is the stationary segment of the motor. The stator has coils of wire that can produce magnetic fields; these coils are organized into groups called phases.
- **Winding phases:** Stepper motors can be either bipolar or unipolar, with bipolar stepper motors having two winding phases, and unipolar stepper motors having four winding phases. Each phase is associated with one winding on the stator.
- **Pulses and control:** A sequence of electrical pulses must be sent to the winding phases to rotate the stepper motor, with the order and timing of these pulses being used to determine the direction and distance of each step.

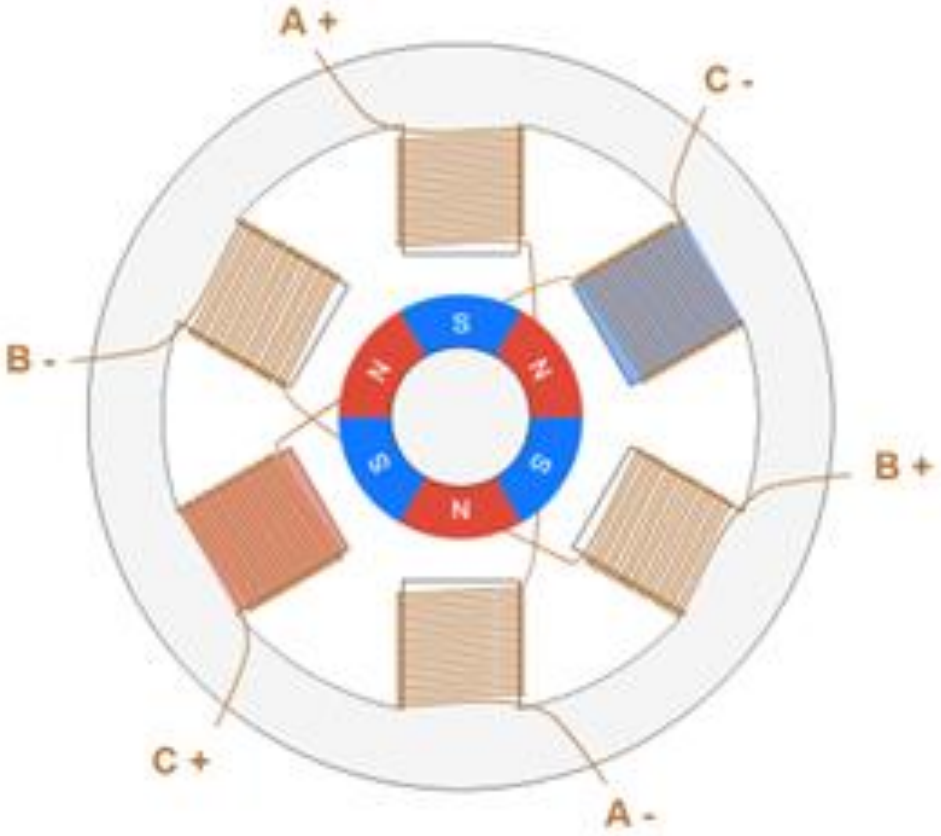


Figure : Stepper Motor

Stepper motors are highly precise because they move in discrete steps. In addition, they were designed to maximize their holding torque, which requires them to maintain a maximum current and makes them ideal for position-holding tasks, such as robotics and camera gimbals.

The basic working principle of the stepper motor is as following:

- By energizing one or more of the stator phases, a magnetic field is generated by the current flowing in the coil and the rotor aligns with this field.
- By supplying different phases in sequence, the rotor can be rotated by a specific amount to reach the desired final position.
- **Figure 2** shows a representation of the working principle. At the beginning, coil A is energized and the rotor is aligned with the magnetic field it produces.
- When coil B is energized, the rotor rotates clockwise by 60° to align with the new magnetic field.
- The same happens when coil C is energized. In the pictures, the colors of the stator teeth indicate the direction of the magnetic field generated by the stator winding.

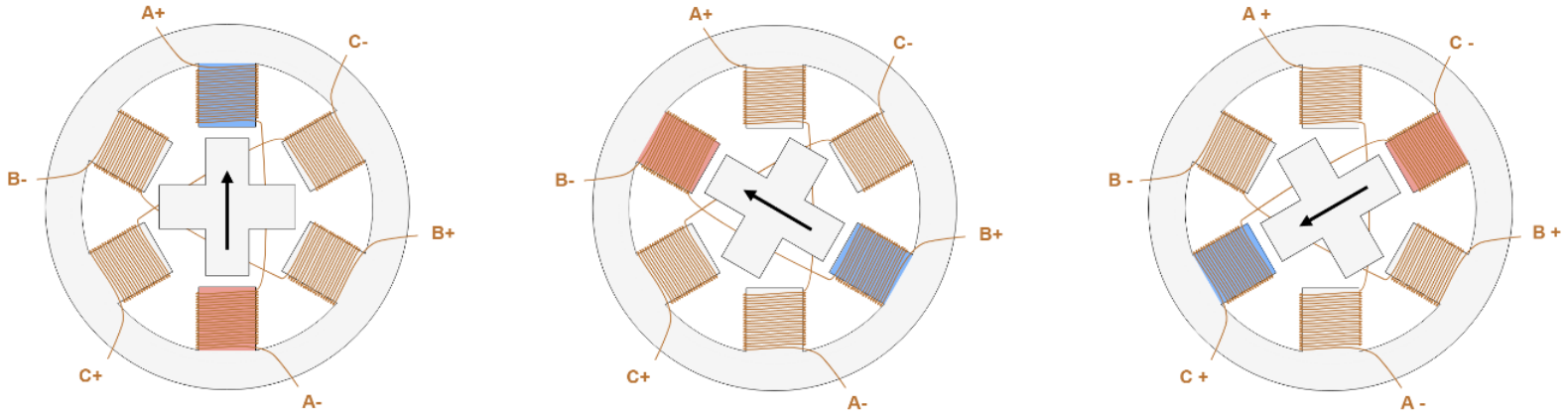


Figure 2: Stepper Motor Steps

Unique Features & Advantages of Stepper Motors

- **Brushless** – Stepper motors are brushless, which means they have a high reliability with little maintenance. The commutator and brushes of conventional motors are some of the most failure-prone components, and they create electrical arcs that are undesirable or dangerous in some environments.
- **Load Independent** – Stepper motors will turn at a set speed regardless of load as long as the load does not exceed the torque rating for the motor.
- **Open Loop Operation** – Stepper motors move in quantified increments or steps, providing exceptional positioning. As long as the motor runs within its torque specification, the position of the shaft is known at all times without the need for a feedback mechanism.
- **Holding Torque** – Stepper motors are able to hold the shaft stationary.
- **Excellent response** to start-up, stopping and reverse.

Difference Between Stepper Motors and Servo Motors

Stepper Motor	Servo Motor
Open loop control as position is calculated by step counts.	Closed loop control using feedback for higher accuracy.
Holds position without any power.	Requires continuous power for holding position using feedback.
Less accurate at high speeds due to delay between pulses.	More accurate at all speeds using feedback control.
Makes a full rotation in discrete steps.	Can rotate continuously in any position.
Less complex driver required.	Require more complex drivers including an amplifier.
Applications include printers, scanners, CNC machines.	Applications include robotics, industrial automation, 3D printers.

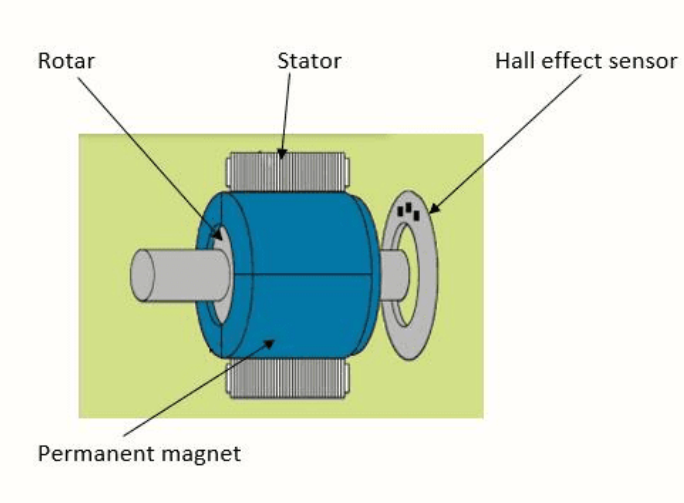
Brushless DC motors (BLDC)

Brushless DC motors (BLDC) are widely used in various applications such as robotics, electric vehicles, drones, and industrial automation systems. Unlike conventional DC motors, BLDC motors do not have brushes, which makes them more reliable and efficient. **Brushless DC motors are also known** as electronically commutated motors (ECMs, EC motors).

BLDC motor is a permanent magnet synchronous electric motor which is driven by direct current (DC) electricity and it accomplishes electronically controlled commutation system (commutation is the process of producing rotational torque in the motor by changing phase currents through it at appropriate times) instead of a mechanically commutation system. BLDC motors are also referred to as trapezoidal permanent magnet motors.

BLDC Motor Components and Construction:

A BLDC motor has three main components: the stator, the rotor, and the Hall effect Sensor.



Stator: The stator is the stationary part of the motor that contains the windings. These windings are made up of insulated copper wire and are arranged in a specific pattern. The stator provides a magnetic field that interacts with the rotor to produce torque.

Rotor: The rotor is the rotating part of the motor that contains permanent magnets. The magnets are arranged in a specific pattern, opposite to that of the stator. The interaction between the magnetic fields of the stator and the rotor produces rotational movement.

Hall effect Sensor or Electronic controller: The Hall effect Sensor is the brain of the motor. It is responsible for controlling the flow of current to the motor windings. The controller also senses the position of the rotor and adjusts the current accordingly to ensure smooth and efficient operation.

Working Principle of BLDC Motor: The working principle of BLDC motors is based on the interaction between the magnetic fields of the stator and the rotor. The stator produces a rotating magnetic field, which interacts with the permanent magnets on the rotor, producing a torque that causes the rotor to rotate.

The Hall Effect Sensor plays a crucial role in the operation of the motor. It controls the flow of current to the motor windings based on the position of the rotor. The controller senses the position of the rotor using sensors or Hall effect devices mounted on the stator. These sensors detect the position of the magnets on the rotor and send signals to the controller.

Based on the signals from the sensors, the controller adjusts the flow of current to the motor windings to ensure that the magnetic fields of the stator and rotor are properly aligned. This ensures that the motor operates efficiently and smoothly, without any loss of power or vibration.

Advantages of BLDC Motors: BLDC motors offer several advantages over conventional DC motors. Some of the advantages are:

Higher efficiency: BLDC motors are more efficient than conventional DC motors due to the absence of brushes. This results in less friction and lower power loss.

Higher power density: BLDC motors have a higher power density compared to conventional DC motors. This means that they can produce more power in a smaller size.

Longer lifespan: The absence of brushes in BLDC motors results in less wear and tear, making them more reliable and durable.

Low maintenance: BLDC motors require less maintenance compared to conventional DC motors. This results in lower maintenance costs and longer service life.

Applications of Brushless DC motor

Brushless DC motors (BLDC) use for a wide variety of application requirements such as

1. Computer hard drives and DVD/CD players
2. Electric vehicles, hybrid vehicles, and electric bicycles
3. Industrial robots, CNC machine tools, and simple belt driven systems
4. Washing machines, compressors and dryers
5. Fans, pumps and blowers.

Microactuator

- A microscopic servomechanism used to supply & transmit a measured amount of energy for the system or another mechanism operation is known as a microactuator.
- Like a general actuator, a microactuator has to meet these standards like fast switching, large travel, high precision, less power consumption, etc.
- These actuators are available in different sizes which vary from millimeters to micrometers, but once they are packaged they can achieve the whole size in centimeters.

- **Microactuator Types**

Microactuators are available in different types which are discussed below.

1. Thermal Microactuator

- A thermal microactuator is a standard component that is used in Microsystems.
- These components are electrically powered through Joule heating otherwise optically activated by using a laser.
- These actuators are used in MEMS designs which include nanopositioners & optical switches.
- The main benefits of thermal microactuators mainly include less operating voltages, high generation of force, and less vulnerability to adhesion failures as compared to electrostatic actuators.
- These actuators need more power & their switching speeds are limited through cooling times.

2. MEMS Microactuator

- MEMS microactuator is one kind of Micro Electro Mechanical System and its main function is to change the energy into motion.
- These actuators combine electrical & mechanical components with micrometer dimensions.
- So, the typical motions attained by these actuators are micrometers.
- MEMS microactuators are mainly used in different applications like ultrasonic emitters, optical beam deflection micromirrors & camera focus systems.
- So these types of microactuators are mainly used to produce a controlled deflection.

3. **Electrostatic Micro Actuator**

- A microactuator driving units which are driven through electrostatic force is known as an electrostatic microactuator.
- The electrostatic microactuator is becoming the most significant building block within computing systems & optical signal processing because of its high density, small size, low power consumption & high speed.
- In general, the operation principle within these systems can be explained as electrostatic attractive energy causing a mechanical revolution, conversion or mirror plate deformation, controlling the phase, power, or light beam direction when it transmits throughout some free space or medium.

4. Piezoelectric Micro Actuator

- Piezoelectric microactuators are very famous and most frequently used in different fields.
- These are designed by mounting piezoelectric elements on top of each other.
- Once a voltage is given to both sides of these elements, then they can expand.
- But it has a complicated structure so it is complex to assemble.
- Piezoelectric micro-actuator is used in different servo control systems to provide ultra-precise positioning & compensation with the potential.

Advantages and Disadvantages

The **advantages of microactuators** include the following.

- The thermal microactuators' benefits are less operating voltages, generation of force is high, and less susceptibility to adhesion failures when compared to electrostatic actuators.
- The microactuators are available in a smaller size, with less power consumption & faster response system.

The **disadvantages of microactuators** include the following.

- Thermal microactuators need more power.
- The switching speed of thermal microactuators is limited by cooling times

Microactuator Applications

The applications of microactuators include the following.

- Microactuator is a small active device used to produce mechanical motion of fluids/ solids. Here motion is produced by changing one form of energy to another form.
- Microactuators are applicable in microfluidics for Lab-on-a-Chip & Implantable Drug Delivery Systems.
- It is a microscopic servomechanism that transmits & supplies a measured amount of energy for another system/mechanism operation.
- Microactuators are used for building small mirrors for projectors & displays.
- MEMS microactuators are mainly used in different applications like ultrasonic emitters, camera focus systems & optical beam deflection micromirrors.
- The force produced by an electric microactuator is mainly utilized to generate mechanical deformations within the material of interest.

Drive Selection

When selecting a motor or drive, consider the following key aspects:

Function: DC motors offer strong starting torques and adaptability, ideal for high-torque needs, while AC motors excel in high-speed processes and low-load operations with easier control of multiple motors.

Location: Motor size, accessibility, and durability are crucial. AC motors require less maintenance, and the environment (temperature, moisture, dust) may necessitate protective enclosures.

Compatibility: Ensure compatibility between power sources, voltage, and horsepower, as well as system software and firmware.

Sustainability: Consider lifespan, maintenance needs, and software update flexibility. AC drives are often more sustainable due to energy efficiency and ease of updates.

Cost: DC motors are cheaper initially, but AC motors save on energy over time. Factor in long-term maintenance and operational savings when considering upfront costs.

- Power transmission systems in robots are essential for transferring energy from the robot's power source (typically motors) to its mechanical parts. There are several types of power transmission systems commonly used in robotics:

1. Gears

- **Function**: Gears transfer motion and torque between components, usually changing speed and direction.
- **Types**:
  - **Spur Gears**: Transfer motion between parallel shafts.
  - **Bevel Gears**: Transfer motion between perpendicular shafts.
  - **Worm Gears**: Provide large speed reductions between perpendicular shafts.



- **2. Belts and Pulleys**
- **Function**: Belts transfer power between two or more pulleys to move various components.
- **Use**: They are quieter and smoother than gears, ideal for longer distances and lightweight loads.





3. Chains and Sprockets

- **Function**: Chains and sprockets work similarly to belts and pulleys but can handle heavier loads and provide more precision.
- **Use**: They are used in heavy-duty applications like industrial robots.





4. Lead Screws and Ball Screws

- **Function**: Convert rotational motion into linear motion.
- **Use**: These are used for precise movement in robotic arms and CNC machines, providing high accuracy.



5. Hydraulic Systems

- **Function**: Use pressurized fluid to transfer power.
- **Use**: Provide high force and are used in large robots and heavy-duty tasks like construction robots.

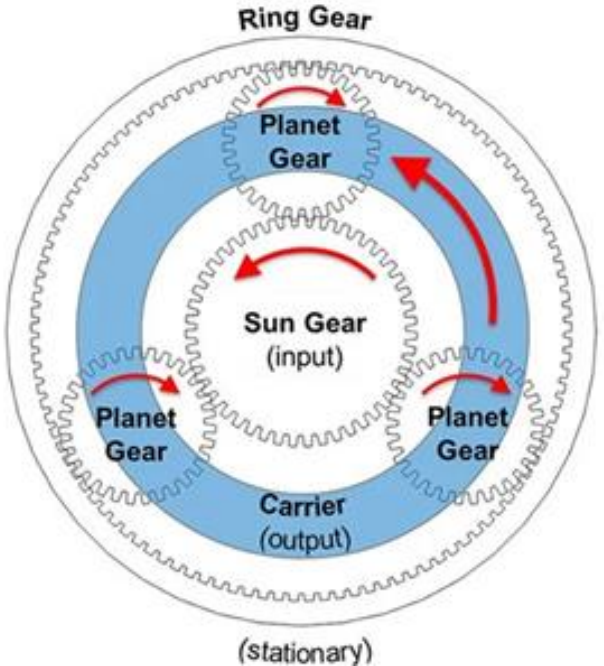
6. Pneumatic Systems

- **Function**: Use compressed air to power actuators.
- **Use**: Common in lightweight, fast-moving robots, such as pick-and-place systems.

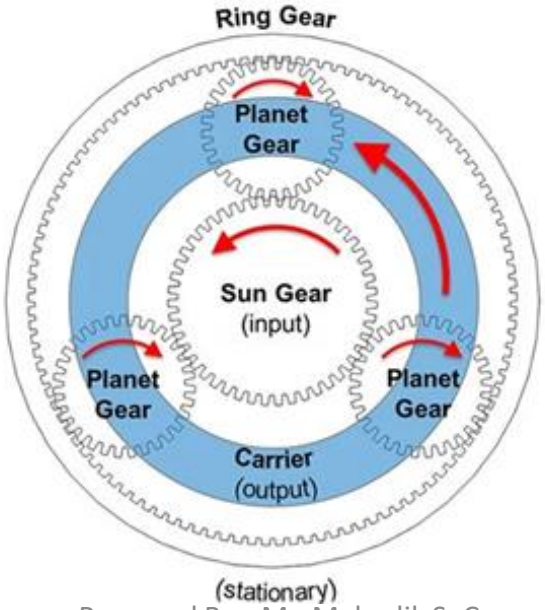
Motion conversion Devices

- A planetary gear is a type of epicyclic gear system comprised of spur gears.
- In planetary gearing (also known as epicyclic gearing), a center gear, called a sun gear, serves as the input and driver of the set. Three or more “driven” gears (referred to as planets) rotate around the sun gear within a planetary gearbox.
- Finally, the planets engage with a ring gear from the inside, which makes an internal spur gear design. Because the planet gears are evenly distributed around the sun gear, planetary gear trains are known to be extremely rugged designs.
- Another benefit of a planetary gearset is that it is easy to convert to a different ratio by simply changing out the carrier and sun gear





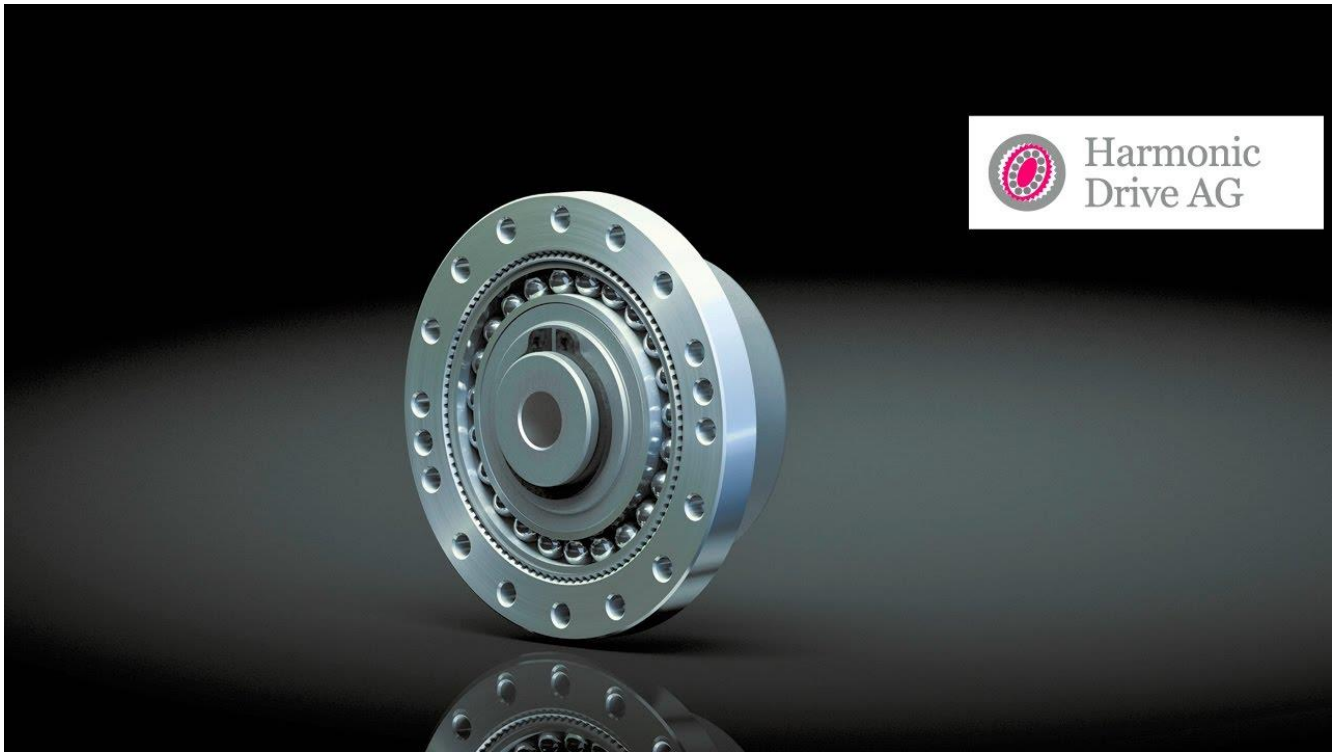
- A planetary gearset uses spur gears that move opposite of each other within the same plane.
- While spur gears are a more basic type of gear in terms of engineering since they do not utilize specialty angles or cuts like bevel or herringbone gearing, they are complex in the tooth shape design.
- Depending on the application, this tooth design will determine where the teeth make contact, which then determines the available power, torque, and speed potential of the gears.



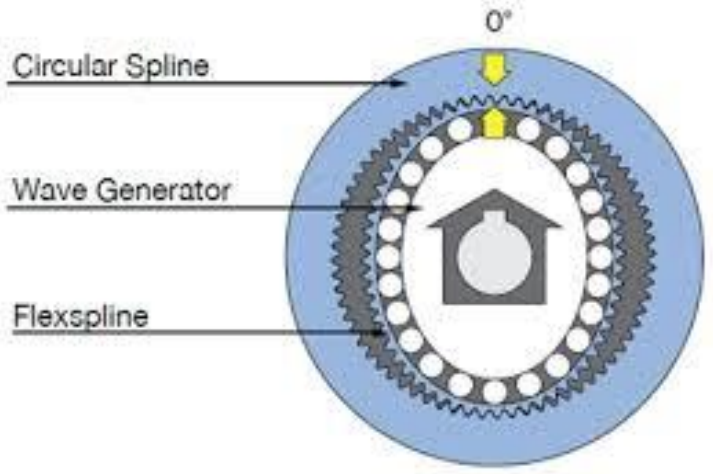
Where is the planetary gearbox usually used (in the transmission):

- In a robot to increase the torque
- In a printing press to reduce the speed of the rollers
- For precise positioning
- In a packaging machine for a reproducible products

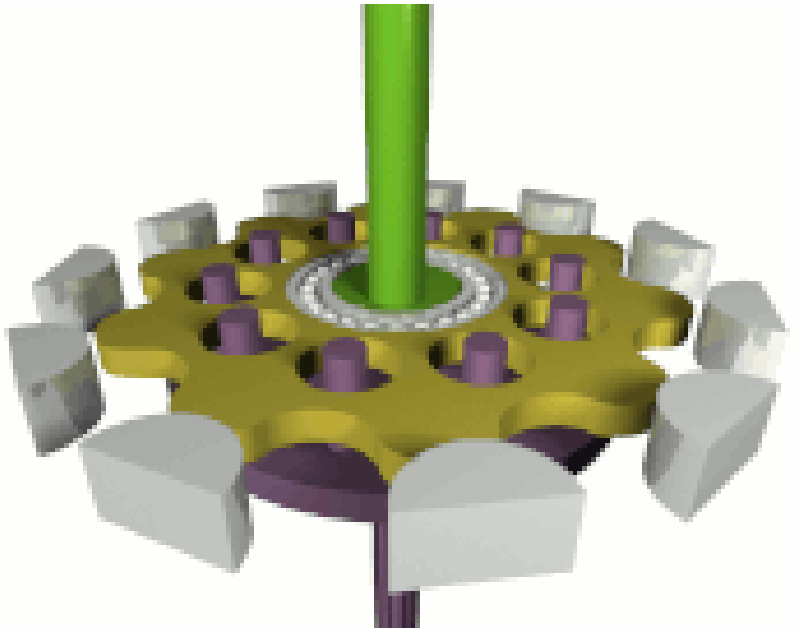
- **Harmonic Drive**, mechanical speed-changing device, invented in the 1950s, that reduces the gear ratio of a rotary machine to increase torque. It operates on a principle different from that of conventional speed changers. The device consists of a thin ring that deflects elastically as it rolls inside a slightly larger rigid circular ring.



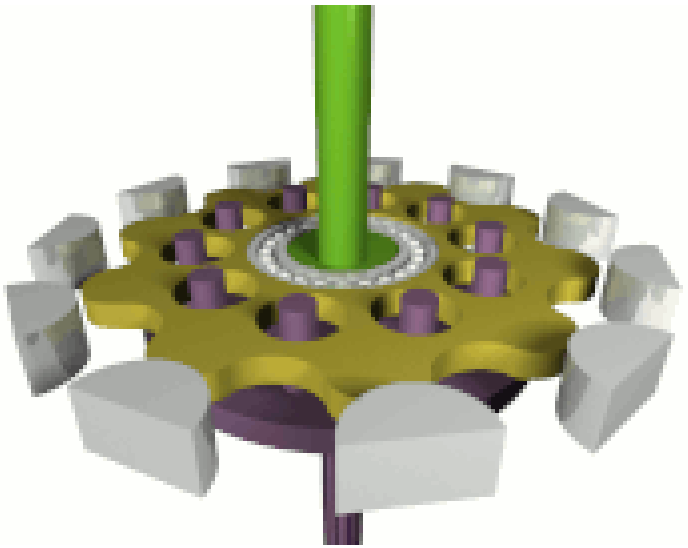
- There are three elements in a basic Harmonic Drive: a circular spline, a flexspline, and a wave generator.
- The circular spline has internal teeth that mesh with external teeth on the flexspline.
- The flexspline has fewer teeth and consequently a smaller effective diameter than the circular spline.
- The wave generator is elliptical in shape and acts as a link with two rollers that rotates within the flexspline, causing it to mesh with the circular spline progressively at diametrically opposite points.
- If the wave generator (the input) rotates clockwise while the circular spline is fixed, the flexspline (the output) will rotate or roll inside the circular spline at a much slower rate in a counterclockwise direction.



- A **cycloidal drive** or cycloidal speed reducer is a mechanism for reducing the speed of an input shaft by a certain ratio. Cycloidal speed reducers are capable of relatively high ratios in compact sizes with very low backlash.



- The input shaft drives an eccentric bearing that in turn drives the cycloidal disc in an eccentric, cycloidal motion.
- The perimeter of this disc is geared to a stationary ring gear and has a series of output shaft pins or rollers placed through the face of the disc.
- These output shaft pins directly drive the output shaft as the cycloidal disc rotates.
- The radial motion of the disc is not translated to the output shaft.



Variable Frequency Drive (VFD)

- AC motor speed is controlled in two ways – either by controlling the voltage or frequency. Frequency controlling gives better control due to constant flux density than voltage control. This is where the working of VFDs comes to play. It is a power conversion device that converts the fixed voltage, fixed frequency of the input power to the variable voltage, the variable frequency output to control AC induction motors.

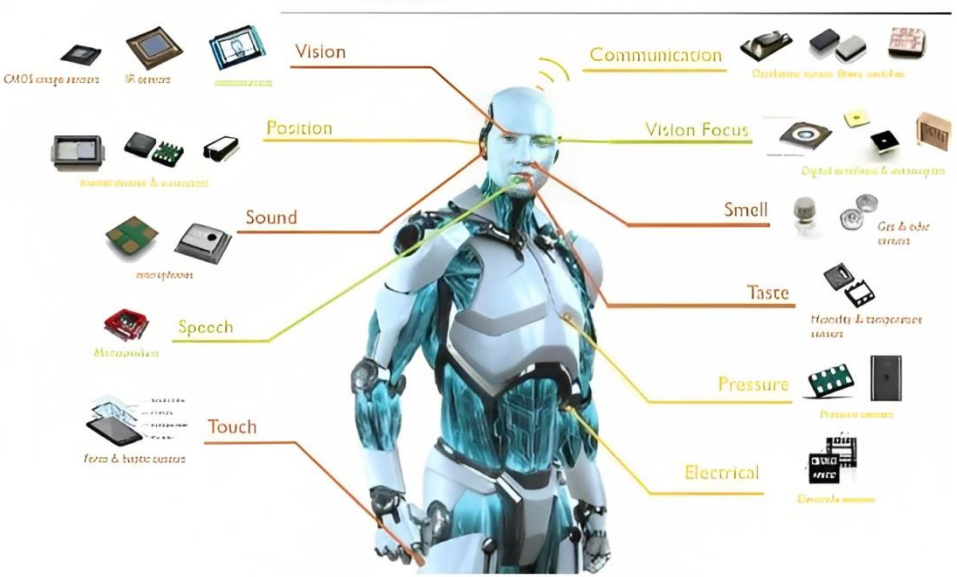
- It consists of power electronic devices (like IGBT, MOSFET), high speed central controlling unit (such as a microprocessor, DSP), and optional sensing devices depending on the application used.

There are many reasons why we may want to adjust this motor speed.

- For example, to Save energy and improve system efficiency
- Convert power in hybridization applications
- Match the speed of the drive to the process requirements
- Match the torque or power of a drive to the process requirements.
- Improve the working environment, Lower noise levels, for example from fans and pumps
- Reduce mechanical stress on machines to extend their lifetime.

Sensors

- A **sensor** is a device that detects and responds to changes in the environment, converting physical signals (such as light, heat, pressure, sound, or motion) into electrical signals.
- These signals are processed by a system, allowing it to react to specific stimuli or gather data.
- In the context of robotics, sensors are used to perceive the robot's surroundings, its own status, or specific conditions, enabling robots to make decisions and adjust their actions accordingly.



- **Role of Sensors in Robotics**
- In robotics, sensors serve as the "eyes," "ears," and "skin" of the robot, playing a critical role in enabling robots to:

1. Perceive the Environment:

  - Sensors help robots sense their environment by gathering real-time data about external conditions such as objects, distances, surfaces, and obstacles. This information is essential for tasks like navigation, object detection, and obstacle avoidance.

- **Example**: An autonomous robot vacuum uses proximity sensors and cameras to detect walls, furniture, or other obstacles, helping it move around efficiently.

2. Provide Feedback for Motion Control:

  - Sensors provide feedback on the robot's position, speed, and orientation. This feedback is essential for maintaining control over the robot's movements, adjusting motor commands, and ensuring the robot follows the intended path.

  Example: A robotic arm uses gyroscopes and encoders to monitor the precise angles and positions of its joints, ensuring it performs tasks with high accuracy.

3. Enable Interaction with Objects:

- Robots use sensors to interact with objects and manipulate them correctly. Sensors detect physical contact, grip strength, pressure, and force applied, ensuring the robot can handle fragile objects without damaging them or apply enough force when needed.
- **Example**: A robotic gripper with force sensors can adjust its grip to gently handle delicate items like fruits without crushing them.

4. Ensure Safety:

- Sensors detect potential hazards or dangerous situations, allowing robots to respond quickly and avoid accidents. This is especially important in collaborative robots (cobots) that work alongside humans in shared environments.
- **Example**: In industrial robots, proximity sensors detect when a human or object enters a dangerous zone, triggering a stop mechanism to prevent accidents.

5. Enable Autonomous Navigation:

- Sensors are vital for enabling robots to navigate autonomously in complex environments. Through data from cameras, LIDAR, ultrasonic sensors, and GPS, robots can create maps, locate themselves within an environment, and plan their movements.
- **Example**: An autonomous car uses LIDAR sensors, cameras, and GPS to map its surroundings, identify obstacles, and safely drive to its destination.

6. Monitor Robot Health:

- Sensors are also used to monitor the internal state of the robot, ensuring that components like motors, joints, and electronics are functioning correctly. Temperature, pressure, and current sensors help detect potential issues like overheating or component failure.
- **Example**: Temperature sensors monitor a robot's motor to prevent it from overheating during prolonged use.

7. Facilitate Human-Robot Interaction:

- Robots equipped with vision sensors, microphones, and tactile sensors can interact with humans through voice commands, gestures, or physical touch. This is particularly useful in service robots, healthcare robots, and social robots.
- **Example**: A service robot can use microphones to understand voice commands and tactile sensors to respond appropriately when touched by a human.

Transducer and Sensors

- A **transducer** is a device that converts one form of energy into another.
- In the context of engineering and electronics, it typically converts physical quantities (such as temperature, pressure, or light) into electrical signals, which can be measured, processed, or controlled by systems like robots or machines.





Key Features of Transducers:

- **Energy Conversion**: A transducer changes physical energy into an electrical signal or vice versa.
- **Types**:
  - **Input Transducers (Sensors)**: Convert physical quantities into electrical signals (e.g., temperature sensors).
  - **Output Transducers (Actuators)**: Convert electrical signals into physical quantities (e.g., loudspeakers convert electrical signals into sound).
- **Example:**

  - A **microphone** is a transducer that converts sound (acoustic energy) into electrical signals.
  - A **speaker** converts electrical signals back into sound.

A **sensor** is a specific type of input transducer that detects and responds to a change in the environment (such as heat, light, motion, or pressure) and converts this into an electrical signal that can be processed or used by a system.

Key Features of Sensors:

- **Measurement**: Sensors measure a specific physical parameter.
- **Sensing Element**: They contain a component (sensing element) that responds to changes in the physical environment.
- **Output**: They provide an electrical signal (current, voltage, or resistance) proportional to the measured quantity.

Example:

- A **temperature sensor** (e.g., thermocouple) measures temperature and converts it into an electrical signal that can be displayed or controlled by a device.

1. Proximity Sensors

Principle: Proximity sensors detect the presence of objects nearby without requiring physical contact. They work by emitting electromagnetic fields or infrared radiation and sensing changes in the return signal.

Types:

- **Inductive Proximity Sensors**: Detect metallic objects using electromagnetic induction.
- **Capacitive Proximity Sensors**: Sense both metallic and non-metallic objects by measuring changes in capacitance.
- **Ultrasonic Proximity Sensors**: Emit ultrasonic sound waves and detect objects based on the time taken for the echo to return.
- **Infrared (IR) Proximity Sensors**: Emit infrared light and detect the reflection from nearby objects.

Applications:

- Detecting objects in automated systems and assembly lines.
- Collision avoidance in robots and vehicles.
- Position detection in conveyor systems.



2. Photoelectric Sensors

Principle: Photoelectric sensors use light to detect the presence of an object. They work by emitting a light beam (typically infrared or visible) and detecting changes in the light path due to an obstruction.

Types:

- **Through-beam**: A light beam travels between an emitter and a receiver, and detection occurs when the beam is interrupted.
- **Retro-reflective**: A single unit contains both the emitter and receiver, with a reflector used to bounce the light back.
- **Diffuse-reflective**: The light emitted from the sensor is reflected off the target back to the receiver in the same housing.

Applications:

- Object detection on production lines.
- Safety systems for machine guarding.
- Counting and sorting objects in packaging machines.



Position Sensors: Piezoelectric Sensor and LVDT

- Position sensors are devices that detect and measure the position, displacement, or movement of an object.
- They play a critical role in industries like automation, robotics, aerospace, automotive, and medical instrumentation.
- Among the various types of position sensors, **Piezoelectric sensors** and **Linear Variable Differential Transformers (LVDT)** are widely used for their high precision and reliability.

1. Piezoelectric Sensors

- The **piezoelectric effect** is the foundation for piezoelectric sensors. Certain materials, such as quartz crystals or specially designed ceramics, generate an electric charge when subjected to mechanical stress or deformation.
- This property can be used to convert mechanical energy (e.g., pressure, strain, or vibration) into electrical energy.
- The amount of electrical charge generated by the material is directly proportional to the applied force, pressure, or displacement, allowing precise measurement.
- Piezoelectric sensors are ideal for dynamic measurements rather than static because they do not generate a continuous electrical signal under constant pressure.



Types of Piezoelectric Sensors

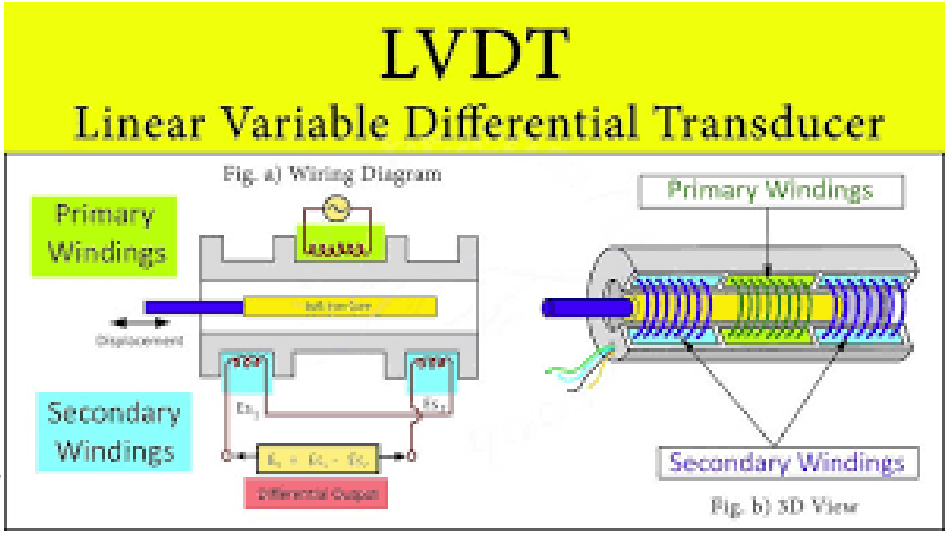
- **Compression Mode**: These sensors generate an electric charge when compressed. They are used in applications where the sensor is placed under a compressive load, such as in vibration or pressure measurements.
- **Shear Mode**: In shear mode sensors, the crystal is stressed by forces parallel to the sensor's surface. These sensors are used in applications where transverse forces are more significant, such as in acceleration or dynamic force measurements.
- **Bending Mode**: These sensors work when bent or deflected. They are more sensitive to small displacements and deformations, making them ideal for applications that require measurement of minute forces or vibrations.

Applications of Piezoelectric Sensors in Robotics

- One of the most important applications of piezoelectric sensors in robotics is force sensing. In robotics, precise control over the force exerted by a robotic arm, gripper, or manipulator is critical to prevent damage to objects or components, especially in delicate tasks like assembling electronics or handling fragile materials.
- Piezoelectric sensors can be used in robots for impact detection and collision avoidance. When a robot interacts with its environment or comes into contact with an object, the piezoelectric sensors detect the impact force or collision, allowing the robot to respond appropriately.
- In micro-robotics, where size and precision are paramount, piezoelectric actuators are often used to achieve fine movements and control. Piezoelectric sensors enable accurate feedback and control in these systems, ensuring precise actuation and positioning.

2. Linear Variable Differential Transformer (LVDT)

- The **Linear Variable Differential Transformer (LVDT)** is an electromechanical sensor used to measure linear displacement or position. It operates based on the principle of **mutual inductance**. The LVDT consists of a primary coil and two secondary coils wound on a cylindrical core. A movable ferromagnetic core, or plunger, is located inside the coils.
- When an AC excitation is applied to the primary coil, it induces a voltage in the secondary coils. The position of the core determines how the magnetic flux couples between the primary and secondary coils. If the core is centered, the induced voltages in the secondary coils are equal and opposite, resulting in zero net output. When the core moves, the differential voltage between the two secondary coils changes, and this differential output is proportional to the displacement of the core.



2.2. Types of LVDTs

- **AC LVDTs**: These sensors use alternating current to power the primary coil and are widely used in industrial applications due to their robustness and ability to operate in harsh environments.
- **DC LVDTs**: These LVDTs incorporate internal signal conditioning circuits, converting the output to a DC signal, making them easier to integrate into systems without the need for external signal conditioning.
- **Digital LVDTs**: These sensors output digital signals instead of analog, simplifying integration with modern control systems and reducing signal noise.

Applications of LVDTs

- **Industrial Automation**: LVDTs are used to measure linear displacement in automated machinery, allowing for precise positioning in tasks such as assembly, packaging, and robotic movement control.
- **Aerospace**: In aircraft systems, LVDTs are used to measure the position of control surfaces, landing gear, and other critical components that require high accuracy and reliability.
- **Power Plants**: LVDTs are used to monitor turbine blade displacement, valve position, and other mechanical components in power generation facilities. They offer durability and can withstand high temperatures and pressures.
- **Material Testing**: LVDTs are used in materials testing to measure deformation and displacement under load, helping engineers determine the mechanical properties of materials.
- **Medical Devices**: LVDTs are employed in medical devices like MRI scanners and robotic surgical instruments where precise, real-time position feedback is essential.
- **Automotive Industry**: LVDTs are used in vehicles for throttle position sensing, suspension systems, and gearbox monitoring to ensure smooth operation and precise control.

Resolvers and Encoders

- **Resolvers** and **Encoders** are two types of position sensors used to detect angular displacement and provide feedback for controlling the motion of rotating machines like motors, robotic systems, and automation equipment.
- Though they serve similar purposes, their underlying operating principles and specific applications differ.



Resolvers

- A **resolver** is an analog sensor that measures the angular position of a shaft. It works on the principle of electromagnetic induction, similar to a rotary transformer. A typical resolver consists of two sets of windings: a primary winding on the stator and two secondary windings on the rotor that are 90 degrees apart. The primary winding generates an alternating current (AC) magnetic field, which induces voltages in the secondary windings. The amplitude of these induced voltages varies based on the angular position of the rotor.
- By analyzing the sine and cosine signals generated by the secondary windings, the exact angular position of the rotor can be determined. The ratio of the two signals (sine and cosine) is used to calculate the angular displacement of the shaft.

Types of Resolvers

- **Brushless Resolver**: This is the most common type, which does not use physical brushes to transfer electrical signals. The absence of brushes makes it more durable and reliable for long-term use in harsh environments.
- **Brushed Resolver**: These resolvers use brushes to transfer electrical signals. Although they are less common today due to wear and tear issues, they are still used in specific legacy systems.
- **Variable Reluctance Resolver**: In this type, the rotor is made of a ferromagnetic material, and the inductance changes as the rotor moves. This change in inductance is used to determine the angular position.

Applications

- **Aerospace and Defense**: Resolvers are used in avionics, missile guidance systems, and radar systems due to their high reliability and ability to function under extreme conditions such as high temperatures, shock, and vibration.
- **Robotics**: Resolvers are often used in robotics for accurate feedback of the robot's joint positions. Their ruggedness makes them suitable for high-precision motion control in robotic arms and automation systems.
- **Electric Motors**: Resolvers are used in motor control systems for electric vehicles (EVs), where they provide feedback on the rotor's position to ensure accurate and efficient motor operation.
- **Industrial Automation**: Resolvers are employed in servo motor feedback systems in CNC machines, packaging machines, and material handling equipment, where precise motion control is necessary.

Encoders

An **encoder** is a digital sensor that converts the angular position or motion of a shaft into an electrical signal. Encoders can be optical, magnetic, or capacitive, depending on the sensing technology used. They produce digital pulses or a binary code corresponding to the shaft's rotational position, speed, or direction.

Encoders are classified into two main categories:

- **Incremental Encoders**: These provide relative position information by generating a series of pulses as the shaft rotates. The number of pulses per revolution (PPR) defines the resolution of the encoder. Incremental encoders cannot determine the absolute position without an external reference point (index pulse).
- **Absolute Encoders**: These provide the exact angular position of the shaft by producing a unique digital code for each position. They can retain position information even if power is lost, making them useful in applications where maintaining position data is critical.

Types of Encoders

- **Optical Encoders**: These encoders use a light source (LED) and a photo detector to detect light passing through a rotating code disk. The disk has patterns of transparent and opaque sections, and as it rotates, the photo detector senses the changing light patterns, which correspond to the shaft's position or speed.
- **Magnetic Encoders**: In magnetic encoders, a magnet attached to the rotating shaft interacts with magnetic sensors, such as Hall effect sensors, to produce position information. These encoders are highly durable and can operate in environments with dust, oil, or moisture.
- **Capacitive Encoders**: These encoders use changes in capacitance between rotating and stationary elements to detect the position of the shaft. They offer high resolution and are less sensitive to environmental factors like temperature and vibration.

Application

- **Robotics**: Encoders are critical in robotic arms and autonomous robots for measuring joint angles, wheel positions, and the speed of moving parts. They enable precise control over movements and ensure accurate positioning.
- **CNC Machines**: In computer numerical control (CNC) machines, encoders provide feedback on the position of cutting tools, ensuring high precision in machining operations.
- **Elevators and Escalators**: Encoders are used to monitor the position and speed of elevator cabins and escalator steps, ensuring smooth and safe operation.
- **Automotive Industry**: Encoders are used in automotive systems to monitor the position of components like the throttle, steering, and wheels. In electric and hybrid vehicles, they are used to control motor position and speed.

Pneumatic Position Sensors

- **Pneumatic position sensors** detect the position of components in a pneumatic system, such as pistons, actuators, or valves, by sensing changes in air pressure or flow.
- They provide non-contact feedback, making them ideal for environments where electrical sensors may be unsuitable due to moisture, dust, or explosive conditions.



Types of Pneumatic Position Sensors:

- **Air-Pilot Sensors**: Detect pressure changes to sense piston position.
- **Pneumatic Proximity Sensors**: Sense proximity of moving parts using air flow or back pressure.
- **Pneumatic Limit Switches**: Trigger air valves to signal position changes.
- **Pressure-Based Sensors**: Detect pressure differentials to determine actuator position.

Applications:

- **Industrial Automation**: Positioning in material handling and assembly lines.
- **Robotics**: Sensing robotic arm positions.
- **Control Systems**: Monitoring valves and actuators.
- **Safety Systems**: Ensuring proper operation in emergency stops.

Range sensors

- **Range sensors** are devices that measure the distance between the sensor and an object. They are widely used in robotics, industrial automation, and mapping systems for precise distance measurements.

Types of Range Sensors:

- **Range Finders**: These sensors measure distance by emitting waves (ultrasonic, infrared, or radio waves) and calculating the time it takes for the signal to return after reflecting off an object (Time of Flight principle).
  - **Applications**: Object detection, obstacle avoidance, level measurement.
- **Laser Range Meters**: These use a laser beam to measure distances with high accuracy. They also work on the Time of Flight principle or by phase-shift detection, where the time or phase difference of the returning laser signal is measured.
  - **Applications**: Surveying, robotics, construction, autonomous vehicles.

Applications of Range Sensors:

- **Robotics**: Used for navigation, obstacle detection, and mapping.
- **Industrial Automation**: For precise positioning, material handling, and quality control.
- **Surveying and Construction**: Laser range meters are used for accurate distance measurement and leveling.

Touch Sensors:

- **Touch sensors** detect physical contact or touch, converting it into an electrical signal. They are commonly used in smartphones, touchscreens, and robotics for user input or object interaction.
- **Types**:
  - **Resistive Touch Sensors**: Respond to pressure, where contact changes electrical resistance.
  - **Capacitive Touch Sensors**: Detect changes in capacitance when a conductive object (like a finger) comes close.
- **Applications**: Touchscreens, robotic grippers, user interfaces.

Force Sensors

- **Force sensors** measure the force applied to an object by converting mechanical stress into an electrical signal. They are used to monitor and control forces in various applications.
- **Types**:
  - **Strain Gauge-Based Sensors**: Use changes in resistance of strain gauges to measure force.
  - **Piezoelectric Force Sensors**: Use piezoelectric materials to generate an electrical charge in response to force.
- **Applications**: Robotics, industrial machinery, medical devices (e.g., prosthetics), automotive testing.

Torque Sensors

- **Torque sensors** measure the rotational force (torque) applied to an object, typically using strain gauges or piezoelectric elements.
- **Types**:
  - **Rotary Torque Sensors**: Measure torque in rotating shafts.
  - **Reaction Torque Sensors**: Measure static torque in non-rotating applications.
- **Applications**: Automotive testing, robotic arms, engine diagnostics, industrial equipment.